

A4

Full-Stack Application

Deployment to the Cloud

Course: Web Applications

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1. Live Application

- **Public URL of frontend**
<https://to-do-app-full-stack-mu.vercel.app/>
- **Functional backend API**
<https://to-do-app-full-stack-bqlr.onrender.com>

2. Source Code Repository

- **Organized project structure**

```
todo-app/  
  frontend/  
    index.html - Complete UI with modals and toasts  
    css/styles.css - Monochromatic, minimalist, blue design  
    js/app.js - Complete CRUD + search + filters  
  backend/  
    index.js - Express server with CORS and error handling  
    package.json - Dependencies (express, mysql2, cors, dotenv)  
    package-lock.json  
    node_modules - node installed modules  
    .env - Environment variables  
    routes/tasks.js - 5 endpoints REST (GET, POST, PUT, PATCH, DELETE)  
    db/  
      connection.js - MySQL connections pool  
      schema.sql - Tasks table  
  docs/  
    architecture.md - Complete documentation in plain text format  
  .gitignore  
  README.md  
  .git
```

- **README.md including:**
 - Project description
 - Setup instructions
 - Deployment explanation[Link to README.md](#)

3. Technical Documentation (PDF or Markdown)

- **Platform selection justification**

Frontend – Vercel

- Automatic deployment on every push to GitHub (`main` branch).
- Optimized for static sites (HTML/CSS/JS) — no build step needed.
- Unlimited free tier with global CDN included.

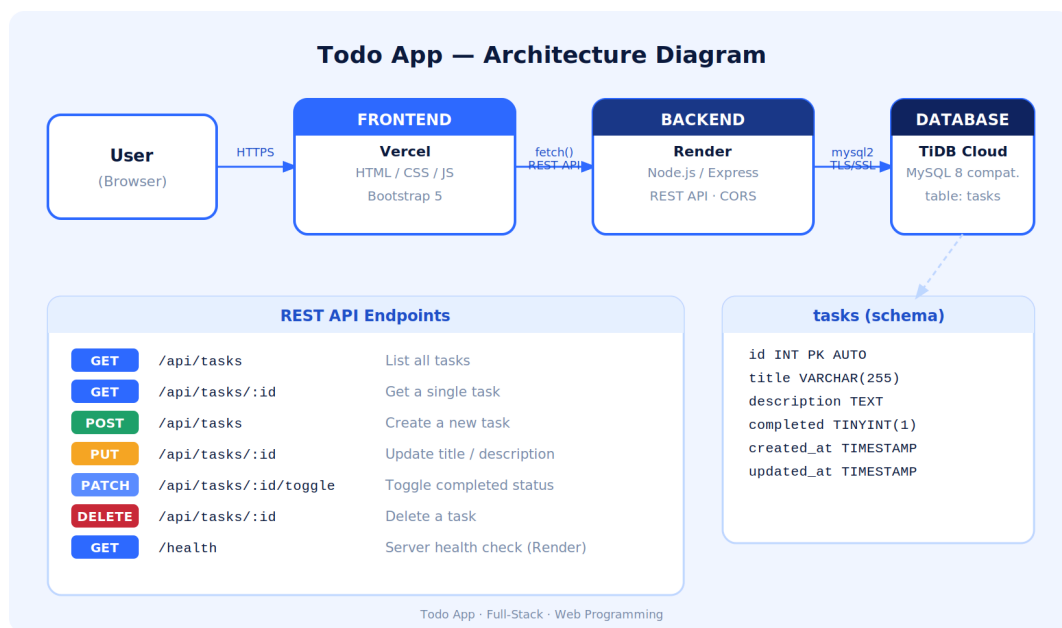
Backend – Render

- Native support for Node.js / Express.
- Environment variables managed directly from the dashboard.
- Automatic deployment from GitHub; zero manual server setup.
- Free tier available (service may take ~30s to wake on first request).

Database – TiDB Cloud

- Fully managed MySQL-compatible (MySQL 8) serverless database.
- No server configuration required; credentials available immediately.
- Supports standard `mysql2` driver — no code changes needed vs. local MySQL.
- Built-in TLS/SSL encryption on all connections.
- Free Serverless tier with 25 GB storage and 250M Request Units/month.
- Advantage over Railway: TiDB Cloud is purpose-built for MySQL workloads, offers better connection pooling, and has a more generous free tier for production use.

● Architecture diagram



● Deployment steps followed

Phase 1 — Local Preparation

1. Project separated into `/frontend` and `/backend`.
2. Verified API and DB work locally with `.env` file.
3. Added `dotenv` support; removed all hardcoded credentials.

Phase 2 — Database (TiDB Cloud)

1. Created account at tidbcloud.com.
2. Created a new **Serverless** cluster (free tier).
3. Copied connection credentials from **Connect** → **General**.
4. Imported `backend/db/schema.sql` via the **SQL Editor**.

Phase 3 — Backend (Render)

1. Created account at render.com.

2. New → Web Service → connected GitHub repository.
3. Root directory: `backend` | Build: `npm install` | Start: `node index.js`.
4. Added all environment variables in the Render dashboard.
5. Deployed and copied the public API URL.

Phase 4 — Connect Backend to TiDB Cloud

1. Confirmed env variables are correct in Render.
2. Verified logs show  MySQL connection established.`

Phase 5 — Frontend (Vercel)

1. Created account at vercel.com.
2. Import Git Repository → selected the repo.
3. Root Directory: `frontend`.
4. Updated `API\_URL` in `js/app.js` with the Render URL.
5. Deployed; confirmed UI loads and API calls succeed.

Phase 6 — Integration Testing

- Create task 
- Read task list 
- Edit title/description 
- Toggle completed 
- Delete task 
- No CORS errors 
- No console errors 

● Challenges faced and solutions

Problem	Cause	Solution
CORS error in production	`FRONTEND_URL` misconfigured	Updated Render env var to exact Vercel URL
TiDB connection refused	Wrong port (3306 instead of 4000)	Updated `DB_PORT=4000` in env vars
Render service "sleeping"	Free tier limitation	First request takes ~30s; health check endpoint helps keep it alive

- Screenshots of:

- Deployment settings

Database:

Connect to Cluster0

Connection Type ⓘ

Public

Branch

main

Database

todo_db

Operating System

Windows (Detected)

Parameters

Connection String

HOST: gateway01.us-east-1.prod.aws.tidbcloud.com

PORT: 4000

USERNAME: Z4nWSCAGJPGAhCu.root

PASSWORD: <PASSWORD>

DATABASE: todo_db

CA: <CA_PATH>

<PASSWORD>: pcJRtkwVSp6azwHB

Backend:

Environment variables.

KEY	VALUE
DB_HOST	gateway01.us-east-1.prod.aws.tidbcloud.com
DB_NAME	todo_db
DB_PASSWORD	pcJRtkwVSp6azwHB
DB_PORT	4000
DB_USER	Z4nWSCAGJPGAhCu.root
FRONTEND_URL	https://to-do-app-full-stack-mu.vercel.app
PORT	10000

WEB SERVICE

to-do-app-full-stack-

Node

Free

Upgrade your instance →

Service ID: srv-d7e1jq7avr4c73edtqrg

CosmicBean-code / to-do-app-full-stack- main

https://to-do-app-full-stack-bqlr.onrender.com

Frontend:

Production Deployment

Repository

Instant Rollback

Visit

Todos las tareas

Tareas

Pendientes

Completadas

Buscar tareas...

C

Comenzar la tarea...

Deployment

to-do-app-full-stack-j37r46gub-cosmicbean-codes-projects.vercelApp

Domains

to-do-app-full-stack-mu.vercelApp

Status

Created

Ready

27m ago by CosmicBean-code

Source

main

f53d203 Documentation update

Static Assets

All (3)

HTML (1)

JS (1)

CSS (1)

/css/styles.css

17 kB

/index.html

7.64 kB

/js/app.js

20 kB

Running application

Tareas

Todas 3

Pendientes 1

Completadas 2

Progreso 67%

Todas las tareas

3 tareas

Buscar tareas...

cosas

12 abr 2026

Bañar a Merita

12 abr 2026

Hacer la documentación

Documentación de To-Do app

12 abr 2026